

The University of Jordan/ Mechanical Engineering Department
Numerical Analysis/ Dr. Hashem Alkhalidi/ Fall Semester 2025-26/ Homework # (1)

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Q1) Use Newton-Raphson method to find the root of the function:

$$f(x) = x^4 - 3x^3 + 3x - 3, \text{ use an initial guess } x_0 = 1.5 \text{ and a tolerance } \varepsilon = 1 \times 10^{-3}$$

Q4) Use Newton-Raphson method to find $\sqrt{55}$

Use a suitable initial guess and do only 5 iterations.

Q5) Use Bisection method to find the root of the function:

$f(x) = \sqrt{x+3} - e^{2x} + 10$, use an initial guesses $x_1 = 0$ & $x_2 = 2$ and a tolerance $\varepsilon = 5 \times 10^{-3}$

Q8) Write the first 4 terms in Taylor series ($x_0 = 2$) for the function:

$f(x) = \ln\sqrt{x^2 + 1}$, and use it to evaluate the value of $\ln\sqrt{5.4}$. Find also the truncation error.

Q9) Write the first 4 non-zero terms in Taylor Maclaurin series ($x_0 = 0$) for the function:

$$f(x) = \tan(x) / x$$

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Q10) Given the following table of matrices:

$A = \begin{bmatrix} 3 & 5 \\ -4 & -2 \end{bmatrix}$	$B = \begin{bmatrix} -5 & 2.5 \\ 0 & -4.5 \end{bmatrix}$	$C = \begin{bmatrix} 3 & -5 & 2 & 1 \\ 2.5 & 6 & -1.5 & -4 \end{bmatrix}$
$D = \begin{bmatrix} 3 & -5 & -2 \\ 1 & 4 & 3 \\ -2 & 3 & -1 \end{bmatrix}$	$E = \begin{bmatrix} -2 & -2.5 & 2 \\ -2 & 0 & 3 \\ 1.5 & 4 & -1 \end{bmatrix}$	$H = \begin{bmatrix} 2 & -3 & -2 & -1 \\ -1 & 4 & 3 & 2 \\ 1 & 4 & 0 & -2 \\ 3 & 3 & 5 & -4 \end{bmatrix}$
$U = \begin{bmatrix} 3 & -4 & 2 & 5 \\ -1 & 4.5 & 3 & 1 \\ 0 & 0 & 1 & 1.5 \\ 0 & 0 & 0 & -2 \end{bmatrix}$	$y = \begin{bmatrix} -4 \\ 6 \\ 7.5 \\ 2 \end{bmatrix}$	$b = \begin{bmatrix} 10 \\ -15 \\ 6 \end{bmatrix}$

Find the following combination of matrices and determinants:

1) $A + 2B - A^T + 14A^{-1}$	2) $\det(D)$
3) $C - A$	4) $\det(E)$
5) $D * E$	6) $\det(H)$
7) $2E * D$	8) $\det(U)$
9) $H * U$	10) D^{-1}
11) $C * H$	12) E^{-1}
13) $D * b$	14) $U^T * y$
15) $C^T * B$	16) $45B^{-1} * (A + 2B^T)$